```
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# CS AI - D
# Roll No. 2
# Subject : Machine Learning
# PRN : 12410541
```

```
import numpy as np
import pandas as pd
```

```
df_csv = pd.read_csv("/content/sample_data/california_housing_test.csv")
df_csv
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
0	-122.05	37.37	27.0	3885.0	661.0	1537.0	606.0	6.6085	344700.0
1	-118.30	34.26	43.0	1510.0	310.0	809.0	277.0	3.5990	176500.0
2	-117.81	33.78	27.0	3589.0	507.0	1484.0	495.0	5.7934	270500.0
3	-118.36	33.82	28.0	67.0	15.0	49.0	11.0	6.1359	330000.0
4	-119.67	36.33	19.0	1241.0	244.0	850.0	237.0	2.9375	81700.0
...	...	...	...	...	...	...	...	...	...
2995	-119.86	34.42	23.0	1450.0	642.0	1258.0	607.0	1.1790	225000.0
2996	-118.14	34.06	27.0	5257.0	1082.0	3496.0	1036.0	3.3906	237200.0
2997	-119.70	36.30	10.0	956.0	201.0	693.0	220.0	2.2895	62000.0
2998	-117.12	34.10	40.0	96.0	14.0	46.0	14.0	3.2708	162500.0
2999	-119.63	34.42	42.0	1765.0	263.0	753.0	260.0	8.5608	500001.0

3000 rows × 9 columns

Next steps: New interactive sheet

```
df_csv.shape
```

```
(3000, 9)
```

```
In [3]: s = pd.Series([1, 3, 5, np.nan, 6, 8])

In [4]: s
```

	0
0	1.0
1	3.0
2	5.0
3	NaN
4	6.0
5	8.0

dtype: float64

```
df_csv.isnull().values.any()  #for checking null values
```

```
np.False_
```

```
df_csv.dtypes #for checking data types
```

	0
longitude	float64
latitude	float64
housing_median_age	float64
total_rooms	float64
total_bedrooms	float64
population	float64
households	float64
median_income	float64
median_house_value	float64

dtype: object

```
(df_csv == 0).sum() #for finding out zero's
```

	0
longitude	0
latitude	0
housing_median_age	0
total_rooms	0
total_bedrooms	0
population	0
households	0
median_income	0
median_house_value	0

dtype: int64

```
df_csv['total_bedrooms'] #to select a single column
```

	total_bedrooms
0	661.0
1	310.0
2	507.0
3	15.0
4	244.0
...	...
2995	642.0
2996	1082.0
2997	201.0
2998	14.0
2999	263.0

3000 rows × 1 columns

dtype: float64

```
df_csv[['total_bedrooms','total_rooms']] #for selecting multiple columns
```

	total_bedrooms	total_rooms
0	661.0	3885.0
1	310.0	1510.0
2	507.0	3589.0
3	15.0	67.0
4	244.0	1241.0
...	...	...
2995	642.0	1450.0
2996	1082.0	5257.0
2997	201.0	956.0
2998	14.0	96.0
2999	263.0	1765.0

3000 rows × 2 columns

```
df_csv.iloc[1:6]
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
1	-118.30	34.26	43.0	1510.0	310.0	809.0	277.0	3.5990	176500.0
2	-117.81	33.78	27.0	3589.0	507.0	1484.0	495.0	5.7934	270500.0
3	-118.36	33.82	28.0	67.0	15.0	49.0	11.0	6.1359	330000.0
4	-119.67	36.33	19.0	1241.0	244.0	850.0	237.0	2.9375	81700.0
5	-119.56	36.51	37.0	1018.0	213.0	663.0	204.0	1.6635	67000.0

```
#sorting data

df_csv.sort_values(by='median_income', ascending=False)
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
2199	-118.20	34.19	38.0	2176.0	266.0	798.0	243.0	15.0001	500001.0
2583	-118.41	34.09	37.0	2716.0	302.0	809.0	291.0	15.0001	500001.0
1600	-118.10	34.13	47.0	2234.0	276.0	749.0	260.0	15.0001	500001.0
1383	-118.37	34.10	37.0	407.0	67.0	100.0	47.0	15.0001	500001.0
161	-117.85	33.62	13.0	5192.0	658.0	1865.0	662.0	15.0001	500001.0
...	...	...	...	...	...	...	...	...	...
2841	-118.27	33.96	38.0	1126.0	270.0	999.0	265.0	0.5495	91700.0
2640	-114.62	33.62	26.0	18.0	3.0	5.0	3.0	0.5360	275000.0
2879	-121.84	38.02	46.0	66.0	22.0	37.0	21.0	0.5360	87500.0
641	-121.04	37.67	16.0	19.0	19.0	166.0	9.0	0.5360	162500.0
185	-118.28	34.02	29.0	515.0	229.0	2690.0	217.0	0.4999	500001.0

3000 rows × 9 columns

```
df_csv.describe()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
count	3000.000000	3000.00000	3000.000000	3000.000000	3000.000000	3000.000000	3000.00000	3000.000000	3000.00000
mean	-119.589200	35.63539	28.845333	2599.578667	529.950667	1402.798667	489.91200	3.807272	205846.27500
std	1.994936	2.12967	12.555396	2155.593332	415.654368	1030.543012	365.42271	1.854512	113119.68747
min	-124.180000	32.56000	1.000000	6.000000	2.000000	5.000000	2.00000	0.499900	22500.00000
25%	-121.810000	33.93000	18.000000	1401.000000	291.000000	780.000000	273.00000	2.544000	121200.00000
50%	-118.485000	34.27000	29.000000	2106.000000	437.000000	1155.000000	409.50000	3.487150	177650.00000
75%	-118.020000	37.69000	37.000000	3129.000000	636.000000	1742.750000	597.25000	4.656475	263975.00000
max	-114.490000	41.92000	52.000000	30450.000000	5419.000000	11935.000000	4930.00000	15.000100	500001.00000

```
df_csv.describe(include='all')
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
count	3000.000000	3000.00000	3000.000000	3000.000000	3000.000000	3000.000000	3000.00000	3000.000000	3000.00000
mean	-119.589200	35.63539	28.845333	2599.578667	529.950667	1402.798667	489.91200	3.807272	205846.27500
std	1.994936	2.12967	12.555396	2155.593332	415.654368	1030.543012	365.42271	1.854512	113119.68747
min	-124.180000	32.56000	1.000000	6.000000	2.000000	5.000000	2.00000	0.499900	22500.00000
25%	-121.810000	33.93000	18.000000	1401.000000	291.000000	780.000000	273.00000	2.544000	121200.00000
50%	-118.485000	34.27000	29.000000	2106.000000	437.000000	1155.000000	409.50000	3.487150	177650.00000
75%	-118.020000	37.69000	37.000000	3129.000000	636.000000	1742.750000	597.25000	4.656475	263975.00000
max	-114.490000	41.92000	52.000000	30450.000000	5419.000000	11935.000000	4930.00000	15.000100	500001.00000

```
df_csv.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   longitude           3000 non-null   float64
 1   latitude            3000 non-null   float64
 2   housing_median_age  3000 non-null   float64
 3   total_rooms         3000 non-null   float64
 4   total_bedrooms      3000 non-null   float64
 5   population          3000 non-null   float64
 6   households          3000 non-null   float64
 7   median_income       3000 non-null   float64
 8   median_house_value  3000 non-null   float64
dtypes: float64(9)
memory usage: 211.1 KB
```

```
df_csv.nunique()
```

	0
longitude	607
latitude	587
housing_median_age	52
total_rooms	2215
total_bedrooms	1055
population	1802
households	1026
median_income	2578
median_house_value	1784

dtype: int64

```
df_csv['median_income'].value_counts()
```

median_income	count
15.0001	9
4.0000	8
3.3750	8
3.2500	7
3.8750	7
...	...
3.2440	1
5.0659	1
5.2449	1
5.0406	1
1.9963	1

2578 rows × 1 columns

dtype: int64

```
df_csv.applymap(type)
```

```
/tmp/ipython-input-3828505095.py:1: FutureWarning: DataFrame.applymap has been deprecated. Use DataFrame.map instead.
  df_csv.applymap(type)
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
0	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>
1	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>
2	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>	<class 'float'>

```
df_csv['housing_median_age'] = df_csv['housing_median_age'].astype(int)
```